

Enhancing Delay Propagation Modeling Using Environmental and Operational Data

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Brent Brakenhoff
brakenhoff.brent@gmail.com
University of Amsterdam
Amsterdam, The Netherlands

Dr. A.M.M. Alsahag, PhD
a.m.m.alsahag@uva.nl
University of Amsterdam
Amsterdam, The Netherlands

1 INTRODUCTION

Train delays tend to occur quite frequently in the Dutch railway system, using machine learning models, the aim of this project is to identify the causes and therefore help prevent train delays. Although there has been research.

1.1 Objective

Develop a comprehensive model to predict and mitigate delay propagation in the Dutch railway system by integrating environmental factors (e.g., weather) and operational data (e.g., maintenance schedules, occupancy levels) alongside network topology.

1.2 Research Motivation

Complex Delay Causes: While the previous thesis focused on topological features, delays often result from multiple interacting factors.

Real-World Relevance:

Integrating diverse data sources makes the model more robust and applicable to real-world railway operations.

1.3 Research Question

How can environmental and operational factors be used to improve the prediction of delay propagation on the Dutch railway system?

- How do environmental factors (e.g., weather, temperature) influence delay propagation in the Dutch railway system?
- What role do operational factors (e.g., maintenance schedules, staffing levels) play in exacerbating or mitigating delays?
- How can machine learning models incorporating these factors improve on topology-only models in predicting delay propagation?

2 RELATED WORK

This research builds on the foundation of the research conducted on US domestic airways by Lei et al. [1] and the following forecasting studies conducted on the Dutch railway network by Wen et al. and Li et al [3] [2]. This research has been inspired by a previous thesis, which forecasted delay propagation on trajectories solely on topological features.

3 METHODOLOGY

3.1 Data Collection

Firstly, the necessary data needs to be gathered and preprocessed. NS has made delay data on the Dutch railway network publicly available, this data includes the delay time and delay causes from 2019 until present, it also includes maintenance and other disruptions. Secondly the environmental features will be sourced from the Dutch weather dataset provided by KNMI, this includes weather data from different weather stations, for some stations even dating back to 1910. The weather at each train station for each day can thus be accurately derived from the KNMI dataset.

3.2 Exploratory Data Analysis

What trajectories are most affected by disruptions and what are the causes of these disruptions? What trajectories are most affected by weather? What trajectories are most often under maintenance/construction?

3.3 Feature Engineering

Combine environmental and operational data with network topology features (e.g., centrality measures).

3.4 Model Development

Train machine learning models (e.g., XGBoost, Random Forest) and evaluate performance with and without additional features. Alternative: Create a dynamic graph-based framework to identify and predict vulnerable routes within the Dutch railway network, enabling proactive measures to prevent cascading failures.

3.5 Validation

Multiple models will be compared across multiple delay scenarios, once with topological features and once with topological, environmental, and operational features.

4 RISK ASSESSMENT

4.1 Data

There do not seem to be any significant risks with the data as the data has already been used in previous research and is all publicly available.

Write a better introductory statement

Remark: (How) do I cite Merel Kämpfer's thesis if it's not published

75 **4.2 Methodology**

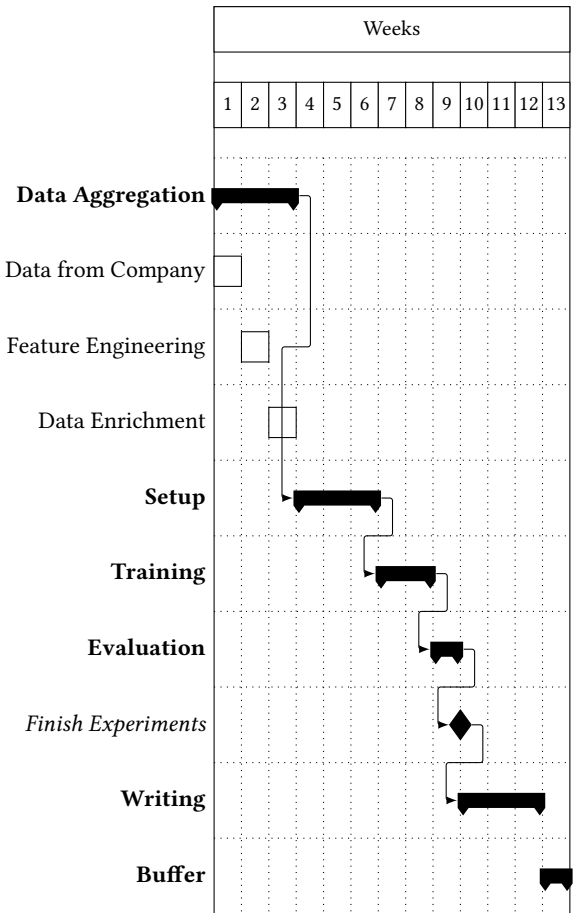
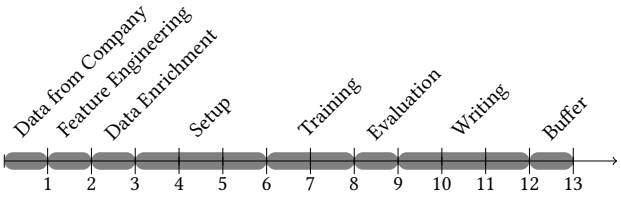
76 As the methodology is based on Lei et al., there should not be any
77 major technical issues or challenges.[1]

78 **4.3 Results**

79 If the results of using environmental and operational features do
80 not seem to or negatively affect accuracy of the models, this project
81 would still highlight disproportionately affected trajectories which
82 is valuable information for improving the Dutch railway system.

83 **5 PROJECT PLAN**

84 Present your timeline here. Finally, you show your academic matu-
85 rity by being able to quantify how much time your work will take
86 (realistically!). Your UvA supervisor must be able to use your visual
87 timeline to check whether you are on schedule. You can either use
88 a timeline as below or a Gantt chart shown on the right.



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